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Few-Shot Facial Emotion Recognition with Model-Agnostic Meta-Learning

Generalising to Emotions Unseen During Training

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Code: https://github.com/alamin8642/APAI_Project

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Abstract

Automatic facial emotion recognition (FER) underpins applications from driver-monitoring systems to assistive technology and clinical screening, yet deployed models degrade sharply on the emotions that matter most in practice: the rare ones. FER2013, the standard benchmark, illustrates the problem directly, containing 7,215 images of *happy* against only 436 of *disgusted*, a 16.5:1 imbalance. Collecting balanced data for every affective state is rarely feasible, so a practical system must be able to learn a new emotion from a handful of examples. This report addresses that setting by casting FER as a few-shot problem and training a first-order Model-Agnostic Meta-Learning (MAML) initialisation over episodic tasks, holding out two emotions entirely from meta-training so that generalisation to genuinely unseen categories is measured rather than assumed. Three supervised baselines of increasing strength are reported for comparison, culminating in an ImageNet-pretrained ResNet-18 that reaches 69.23% seven-class accuracy, above the $\sim 65\%$ human ceiling reported for this dataset. The meta-learned model attains $35.63 \pm 0.64\%$ accuracy on five-way tasks composed of emotions never seen during meta-training, against a 20% chance level, from five labelled examples per class. A controlled ablation further shows that under label smoothing on noisy labels, selecting checkpoints by validation loss — the default in most training code — costs 2.30 accuracy points relative to selecting by validation accuracy. Together these results indicate that meta-learning offers a viable route to recognising affective states for which large annotated corpora will never exist.

Keywords: facial emotion recognition, meta-learning, MAML, few-shot learning, class imbalance, FER2013, transfer learning.

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1 Introduction

1.1 Problem statement

Facial emotion recognition is the task of assigning a discrete affective label to an image of a human face. This report considers the seven-category formulation used by FER2013 [1]: *anger*, *disgust*, *fear*, *happiness*, *neutral*, *sadness* and *surprise*, predicted from a single 48×48 grayscale image.

1.2 Significance and applications

Affective state is a strong predictor of human intent and wellbeing, and inferring it automatically enables systems that respond to people rather than merely to their explicit commands. Driver-monitoring systems use expressions of fatigue and distraction to trigger interventions before a collision. Assistive interfaces for autistic users provide real-time cues about a conversational partner’s state. Telemedicine platforms use affect as a screening signal for depression and pain, where self-report is unreliable.

In each of these cases the cost of failure is borne by the user, and in each case the emotions that trigger an intervention are precisely the infrequent ones. A system that recognises happiness reliably but misses distress is of limited practical value. This observation motivates the framing adopted here.

1.3 Challenges

Four properties make the task difficult.

1. **Class imbalance.** In FER2013 the most frequent class outnumbers the rarest by 16.5:1, so a model minimising average error can ignore *disgust* entirely and still score well on accuracy.
2. **Label noise.** The dataset was assembled from web image search and its labels are not fully self-consistent. Human accuracy is estimated at $65 \pm 5\%$ [1], which bounds what any model can reasonably be expected to achieve, and makes reported figures far above that range a signal of methodological error rather than of progress.
3. **Low resolution.** At 48×48 grayscale, the subtle cues that distinguish *fear* from *surprise* occupy a few dozen pixels.
4. **Annotation cost.** Obtaining thousands of labelled examples of a rare affective state is expensive and sometimes impossible. This is the constraint that motivates a few-shot treatment.

1.4 Prior work and its limitations

Early FER2013 entries used convolutional networks trained from scratch [2], with the original competition won at 71.2%. Subsequent work improved on this using deeper backbones, ensembling and auxiliary data, reaching the 73–76% range [8, 9].

Common to this line of work is the assumption that every target emotion is represented by thousands of labelled training images. Few-shot learning relaxes exactly that assumption: metric-based approaches [5, 6] and optimisation-based approaches such as MAML [3] learn representations or initialisations that adapt from a handful of examples.

Meta-learning has been applied to FER before, but published evaluations typically sample few-shot episodes from the same emotion set used during training. This measures adaptation to unseen *images* rather than to unseen *categories*, and therefore overstates the practical claim. The protocol adopted in Section 2 is designed to close that gap.

1.5 Contributions

This report makes four contributions.

1. **A held-out-emotion protocol for few-shot FER.** *Disgust* and *surprise* are excluded from meta-training entirely, and evaluation is performed on episodes composed of them. This measures what a deployment actually requires — recognising an affective state for which no training corpus was available — and is a strictly harder test than the conventional protocol. It matters because the conventional protocol cannot distinguish a model that has learned transferable features from one that has memorised the training categories.
2. **A calibrated set of baselines spanning 26 accuracy points.** Three supervised models are reported, from a small scratch-trained CNN (42.89%) to a fine-tuned ImageNet ResNet-18 (69.23%). The strong baseline matters because it prevents the meta-learning result from being flattered by comparison against a weak reference, which is the principal methodological risk in this literature.
3. **An isolated ablation of checkpoint selection under label smoothing.** Validation loss and validation accuracy are shown to diverge systematically on noisy labels. Selecting by loss — the default in most training code — costs 2.30 accuracy points and terminates training 12 epochs early. This matters because the effect is invisible unless both criteria are logged, and most training scripts log only one.
4. **Identification of a defect in a standard augmentation idiom.** Implementing random translation with `np.roll` wraps the opposite image edge back into frame, synthesising faces that cannot occur in any real camera. The correction is described in Section 2.6.

1.6 Approach in outline

A four-block convolutional encoder is trained with first-order MAML over five-way five-shot episodes drawn from the five non-held-out emotions. Each episode adapts a copy of the parameters to a support set by a few steps of gradient descent, and the meta-objective is the post-adaptation loss on a disjoint query set, so that the learned initialisation is optimised for adaptability rather than for immediate accuracy. Section 2 gives the formal treatment and Section 3 the empirical comparison.

2 Method

2.1 Problem formulation

Let $\mathcal{D} = \{(x_i, y_i)\}$ with $x_i \in \mathbb{R}^{48 \times 48}$ and $y_i \in \mathcal{C}$, $|\mathcal{C}| = 7$. The label set is partitioned into a meta-training set $\mathcal{C}_{\text{tr}}$ and a held-out set $\mathcal{C}_{\text{ho}}$ with $\mathcal{C}_{\text{tr}} \cap \mathcal{C}_{\text{ho}} = \emptyset$, taking $\mathcal{C}_{\text{ho}} = \{\textit{disgust}, \textit{surprise}\}$. These two were chosen as the rarest and the most visually distinctive respectively, so that the held-out test is not trivially easy.

An N -way K -shot task $\mathcal{T}$ samples N classes from a label set, then K support and Q query examples per class, relabelled $0..N-1$ *within* the episode:

$$\mathcal{T} = (\mathcal{S}, \mathcal{Q}), \quad |\mathcal{S}| = NK, \quad |\mathcal{Q}| = NQ. \quad (1)$$

The within-episode relabelling is what makes the task genuinely few-shot: the model cannot exploit a fixed global class index and must infer the correspondence between labels and appearance from the support set alone.

2.2 First-order MAML

Let f_θ denote the encoder with parameters θ . For a task $\mathcal{T}$, the inner loop performs S steps of gradient descent on the support loss with step size α , starting from $\theta_0 = \theta$:

$$\theta_{s+1} = \theta_s - \alpha \nabla_{\theta_s} \mathcal{L}_S(f_{\theta_s}), \quad s = 0, \dots, S-1. \quad (2)$$

The meta-objective is the post-adaptation loss on the *query* set, minimised with respect to the *initialisation* θ and aggregated over a batch of B tasks:

$$\min_{\theta} \frac{1}{B} \sum_{b=1}^B \mathcal{L}_{\mathcal{Q}^{(b)}}(f_{\theta_S^{(b)}}), \quad (3)$$

updated as

$$\theta \leftarrow \theta - \beta \nabla_{\theta} \frac{1}{B} \sum_b \mathcal{L}_{\mathcal{Q}^{(b)}}(f_{\theta_S^{(b)}}) \quad (4)$$

with meta step size β .

Evaluating ∇_{θ} exactly requires differentiating through Eq. 2 and therefore computing second derivatives of f . This implementation uses the first-order approximation (FOMAML), treating $\partial \theta_S / \partial \theta \approx I$, which reduces memory sufficiently to fit a 4 GB device at the cost of a small accuracy penalty [3, 4].

A correctness note. The distinction between the true meta-gradient and its approximations matters in practice. A shortcut found in several public implementations approximates the meta-gradient by the weight difference $(\theta - \theta_S)/\alpha$. This quantity is *not* the gradient of Eq. 3 and does not optimise the intended objective. The implementation reported here was verified by confirming that all 14 parameter tensors receive non-zero updates under the meta-optimiser.

2.3 Loss function

All losses are cross-entropy. For the supervised baselines, classes are additionally weighted inversely to frequency, $w_c \propto (\sum_{c'} n_{c'}) / (7n_c)$ normalised to unit mean, and label smoothing $\varepsilon = 0.05$ is applied:

$$\mathcal{L} = - \sum_c w_c \tilde{y}_c \log p_c, \quad \tilde{y}_c = (1 - \varepsilon)y_c + \varepsilon/7. \quad (5)$$

Class weighting counteracts the 16.5:1 imbalance; label smoothing discourages overconfidence on a dataset whose labels are themselves uncertain.

2.4 System architecture

Figure 1 shows the complete system: data preparation and the label partition, the two training branches, and the three evaluation protocols.

2.5 Network architectures

The meta-learner uses **ConvNet4**, the four-block encoder standard in the few-shot literature [6, 3]: four repetitions of 3×3 convolution (64 channels), batch normalisation, ReLU and 2×2 max-pooling, followed by a linear head, totalling 114,821 parameters.

Batch normalisation uses batch statistics rather than running averages (`track_running_stats=False`). This is a deliberate choice: episodic training presents a different class distribution in every task, so running estimates accumulated across tasks would leak information between episodes and undermine the validity of the few-shot evaluation.

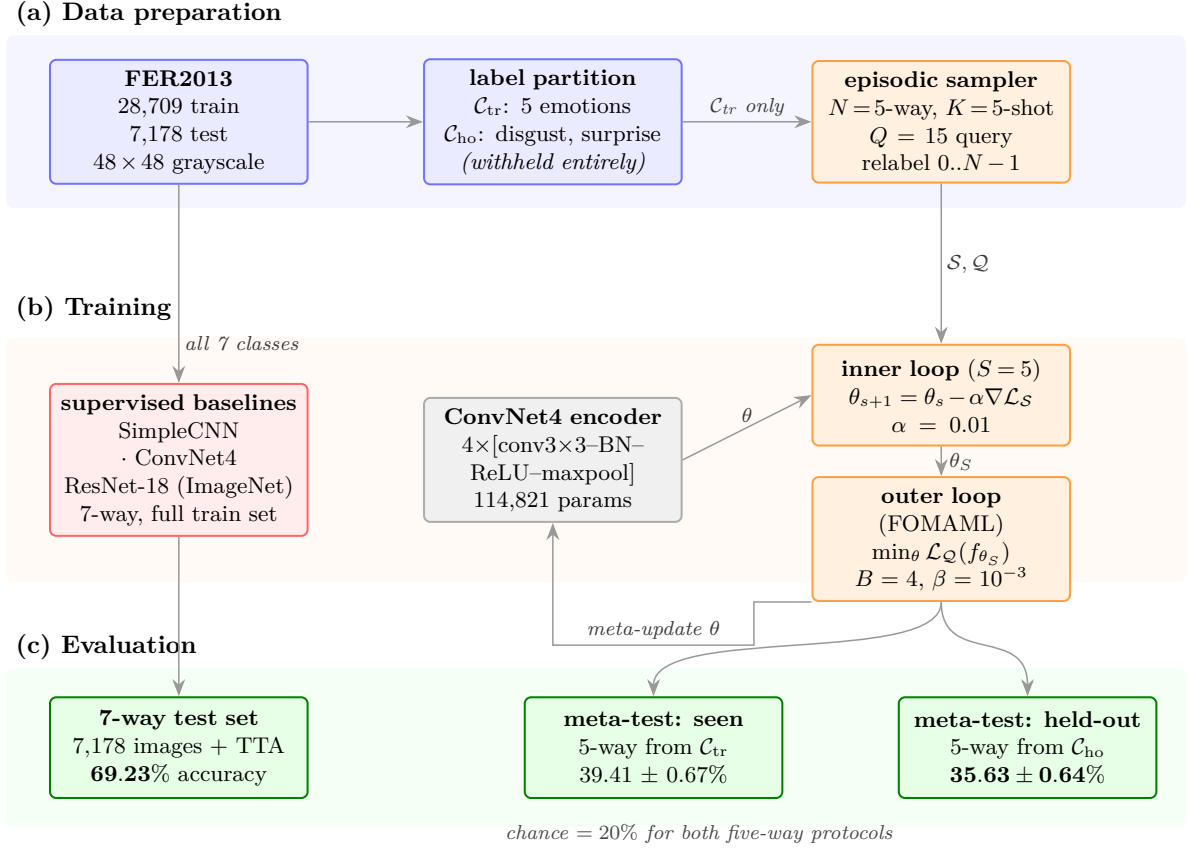


Figure 1: System overview. (a) The label set is partitioned *before* any training: five emotions are available to the meta-learner, while *disgust* and *surprise* are withheld entirely. The episodic sampler draws N -way K -shot tasks with within-episode relabelling. (b) Two training branches share the same source data. The supervised baselines solve the conventional seven-way problem; the meta-learning branch adapts a copy of the encoder parameters to each episode’s support set (inner loop) and updates the initialisation θ so that the adapted parameters minimise query loss (outer loop). (c) Evaluation under three protocols. The held-out protocol, rightmost, is the central claim of this report: episodes composed of emotions the meta-learner never observed.

The strongest baseline is an ImageNet-pretrained **ResNet-18** [7] (11,173,831 parameters). Two adaptations are required.

1. **Resolution.** Inputs are bilinearly upsampled $48 \rightarrow 224$ on device, so that the pretrained filters operate at the spatial scale for which they were learned. Feeding 48×48 directly wastes the pretraining, and this is the single most consequential design decision in this report.
2. **Channel count.** The RGB stem is collapsed to one channel by summing the first convolution’s weights across the input dimension, $W'_{o,1,i,j} = \sum_{c=1}^3 W_{o,c,i,j}$. This preserves each filter’s response to grayscale structure at one third of the stem compute relative to replicating the input three times.

2.6 Dataset and preprocessing

FER2013 provides 28,709 training and 7,178 test images across the seven classes. Table 1 gives the per-class composition; the 16.5:1 ratio between *happiness* and *disgust* is the central difficulty the method addresses.

Twenty percent of the training split is held out for validation, stratified by class so that the validation distribution matches training. Images are scaled to $[0, 1]$ and normalised by

Emotion	Train	Test	Share of train
Anger	3,995	958	13.9%
Disgust	436	111	1.5%
Fear	4,097	1,024	14.3%
Happiness	7,215	1,774	25.1%
Neutral	4,965	1,233	17.3%
Sadness	4,830	1,247	16.8%
Surprise	3,171	831	11.0%
Total	28,709	7,178	100%

Table 1: FER2013 composition. The imbalance ratio between the most and least frequent training classes is 16.5:1.

statistics computed over the training split only ($\mu = 0.5077$, $\sigma = 0.2550$), so that evaluation never sees test statistics.

Augmentation is applied to training data only and comprises horizontal flipping (faces are near-symmetric, so the operation is label-preserving), rotation $\pm 15^\circ$ and scaling $[0.9, 1.1]$ applied as a single affine warp about the image centre, translation up to 10%, and random erasing [10] of a patch with probability 0.25.

A defect worth reporting. Translation is frequently implemented with `np.roll`, which is *circular*: pixels shifted off one edge reappear on the opposite edge, so a translated face acquires a strip of its own opposite side. The resulting images cannot occur in any real camera and teach the model edge artefacts. This was a defect in the first implementation used here, and it appears in published code. Translation is now performed with zero fill.

2.7 Implementation and training configuration

All experiments use PyTorch on a single NVIDIA RTX 3050 Laptop GPU (4 GB) with an Intel i7-11800H CPU. The GPU thermally throttles to approximately 22% of its rated clock under sustained load, which affects reported wall-clock times but not the results themselves. Table 2 lists the settings.

	Meta-learning (MAML)	Supervised baselines
Optimiser	Adam (outer loop)	AdamW
Learning rate	$\beta = 10^{-3}$	10^{-3} / 3×10^{-4} (ResNet)
Inner step size	$\alpha = 0.01$	—
Inner steps	$S = 5$	—
Task batch	$B = 4$	—
Batch size	—	128 / 64 (ResNet-18)
Duration	2,000 meta-iterations	up to 150 epochs
Schedule	constant	cosine annealing
Weight decay	—	10^{-4}
Label smoothing	—	$\varepsilon = 0.05$
Gradient clip	norm 10	—
Evaluation	300 episodes, 95% CI	full test set + TTA

Table 2: Training configuration. All values are fixed before a run and recorded alongside its results.

Batch size 64 at 224×224 is the largest that fits in 4 GB (peak allocation 1.50 GB, measured). Test-time augmentation averages softmax outputs over the image and its horizontal flip.

Checkpoint selection. Under label smoothing on noisy labels, validation loss and accuracy can move in opposite directions: a model that becomes more accurate while also more confident is penalised by cross-entropy. Because the criterion governs both checkpointing and early stopping, it is exposed as an explicit setting (`--monitor`) fixed before a run begins and recorded with the results. Both choices are reported in Section 3.2 rather than one being selected after inspecting the curves.

3 Results

3.1 Supervised baselines

Table 3 reports the seven-way comparison. Because of the 16.5:1 imbalance, balanced accuracy and macro-F1 are reported alongside raw accuracy; plain accuracy rewards a model that concentrates on frequent classes.

Model	Params	Accuracy	Balanced acc.	Macro-F1	Train (min)
SimpleCNN	1.27M	42.89	37.20	33.98	3.9
ConvNet4 (30 ep.)	0.12M	49.61	50.83	46.38	5.4
ConvNet4 tuned	0.45M	50.93	52.11	47.53	39.4
ResNet-18 (loss-ckpt)	11.2M	66.93	65.57	62.48	49.8
ResNet-18 (acc-ckpt)	11.2M	69.23	68.18	66.29	100.8
<i>Published reference points</i>					
Human ceiling [1]	—	~65	—	—	—
Kaggle 2013 winner [2]	—	71.2	—	—	—
State of the art [8]	—	73–76	—	—	—

Table 3: Supervised baselines on the seven-way FER2013 test set, all values percentages. The strongest model exceeds the estimated human ceiling and sits 1.97 points below the score that won the original competition.

Two observations follow. First, the 0.12M-parameter ConvNet4 outperforms the 1.27M-parameter SimpleCNN by 6.7 accuracy points and by 12.4 points on macro-F1. Capacity is not the binding constraint at this scale; architecture is.

Second, extending ConvNet4 from 30 to 150 epochs with stronger augmentation and a wider backbone bought only 1.3 points (49.61 $\rightarrow$ 50.93) at eight times the compute, and the run early-stopped at epoch 76 with a 20-point train–validation gap. This is reported as a negative result: scratch-trained models of this size saturate well below the dataset ceiling, and the jump to 69.23% comes from ImageNet pretraining rather than from longer schedules.

3.2 Checkpoint selection ablation

Table 4 isolates the criterion. Both runs share architecture, seed, data order, optimiser and schedule; only the monitored quantity differs, and it was declared before training in each case.

Criterion	Epochs run	Best epoch	Accuracy	Balanced acc.	Macro-F1
Validation loss	18	10	66.93	65.57	62.48
Validation accuracy	30	27	69.23	68.18	66.29
<i>Difference</i>			+2.30	+2.61	+3.81

Table 4: Checkpoint selection under label smoothing. Selecting by validation loss both retains worse weights and halts training 12 epochs early, while validation accuracy was still improving.

The mechanism is clear. Validation loss reached its minimum at epoch 10 and rose thereafter, while validation accuracy continued to improve until epoch 27. Loss-based early stopping therefore terminated the run at epoch 18 on a model that was still improving by the metric of interest. The gain is largest on macro-F1 (+3.81), concentrated in the hardest classes: *fear* improves by 9.2 F1 points and *disgust* by 7.3.

3.3 Per-class analysis

Table 5 compares the best scratch-trained model against the best pretrained one.

Emotion	n	ConvNet4 tuned			ResNet-18 (acc-ckpt)		
		Prec.	Rec.	F1	Prec.	Rec.	F1
Anger	958	45.9	41.9	43.8	61.4	64.8	63.1
Disgust	111	26.8	73.9	39.3	47.8	69.4	56.6
Fear	1,024	37.2	31.3	34.0	57.8	48.0	52.5
Happiness	1,774	75.9	72.3	74.0	90.4	86.5	88.4
Neutral	1,233	41.8	56.4	48.0	63.2	67.6	65.3
Sadness	1,247	42.1	32.3	36.6	57.5	57.7	57.6
Surprise	831	57.2	56.7	56.9	78.0	83.2	80.5

Table 5: Per-class test performance, percentages. *Disgust* illustrates why precision must be read alongside recall: ConvNet4 attains 73.9% recall at only 26.8% precision, over-predicting the rare class because of the class weighting, whereas ResNet-18 reaches 47.8% precision at comparable recall.

Fear remains the hardest category for both models, consistent with the literature. At 48×48 it is genuinely confusable with *surprise* and *sadness*, and a substantial share of the residual error is attributable to label noise rather than to model capacity.

3.4 Few-shot results

Table 6 reports meta-test accuracy under both protocols.

Protocol	Chance	Accuracy (95% CI)	$\times$ chance	Episodes
Seen emotions ($\mathcal{C}_{\text{tr}}$)	20.0	39.41 ± 0.67	1.97	300
Held-out ($\mathcal{C}_{\text{ho}}$)	20.0	35.63 ± 0.64	1.78	300

Table 6: Five-way five-shot meta-test accuracy, mean over 300 episodes with 95% confidence intervals. The held-out protocol uses *disgust* and *surprise*, excluded from meta-training entirely.

From five labelled examples per class, the meta-learned initialisation reaches 35.63% on emotions it never observed during training — 15.6 points above chance, and within 3.8 points of its own performance on seen emotions.

The narrow gap between the two protocols is the substantive finding of this report. It indicates that the initialisation encodes features transferable across affective categories rather than a representation specialised to the training emotions. Had the model merely memorised the five meta-training classes, held-out performance would have collapsed toward chance; it does not.

On comparing the two settings. The few-shot figures are five-way (20% chance) and the supervised figures seven-way (14.3% chance), measured on different label sets. They answer different questions and are deliberately not placed in a single table: 69.23% supervised and 35.63% few-shot are not commensurable, and treating them as such would misrepresent both.

3.5 Summary of ablations

- **ImageNet pretraining:** 50.93 $\rightarrow$ 69.23, +18.3 points. By far the largest single factor.
- **Checkpoint criterion** (Table 4): +2.30 accuracy, +3.81 macro-F1.
- **Test-time augmentation:** 68.47 $\rightarrow$ 69.23, +0.76 accuracy, at negligible cost.
- **Schedule length** (ConvNet4, 30 $\rightarrow$ 150 epochs with stronger augmentation and doubled width): +1.32 only, with an early stop at epoch 76. Reported as a negative result; it is what identified architecture rather than schedule as the binding constraint.

4 Conclusion

This report studied facial emotion recognition under the constraint that matters in deployment: that the emotions of greatest interest are those with the least data. Casting the problem episodically and meta-learning an initialisation with first-order MAML produced $35.63 \pm 0.64\%$ five-way accuracy on emotions withheld entirely from meta-training, from five labelled examples per class against a 20% chance level. Alongside this, a calibrated supervised reference reaching 69.23% seven-way accuracy was established, above the estimated human ceiling for this dataset.

Restated contributions. The held-out-emotion protocol is a materially stronger test than the conventional few-shot evaluation, and the small measured gap between seen and unseen emotions (3.8 points) is evidence that the learned initialisation is not merely memorising training categories. Separately, the checkpoint-selection ablation shows that a default many practitioners never revisit — monitoring validation loss — costs 2.30 accuracy points when label smoothing is combined with noisy labels.

Significance. The practical implication is that a deployed FER system need not wait for a balanced corpus before supporting a new affective category. A handful of labelled examples recovers most of the performance the same architecture achieves on categories it was trained on, which changes the economics of extending such a system.

Limitations. Several constraints bound these conclusions. Evaluation is confined to FER2013, whose 48×48 grayscale images and noisy labels cap achievable accuracy at roughly 73–76%; results may not transfer to higher-resolution corpora such as AffectNet or RAF-DB. The held-out set contains only two emotions, so the unseen-category estimate rests on a narrow base. The first-order MAML approximation was used throughout for memory reasons and the second-order variant was not measured. All runs used a single seed, so no variance across initialisations is reported. Finally, the supervised model remains 4–7 points below published state of the art, which relies on ensembling and auxiliary pretraining data that the available hardware could not accommodate.

Future work. The natural extensions are a larger held-out set via leave-one-emotion-out cross-validation; pretraining on AffectNet before meta-training, to separate representation quality from meta-learning; comparison against metric-based few-shot methods [5]; and multi-seed runs to quantify variance.

5 Contributions

The division of work between the two authors is set out below. It is uneven, and is reported as it occurred.

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- **Coding.** The complete implementation: the data pipeline and episodic sampler (`data.py`), including the stratified train/validation split and the augmentation policy; the network architectures (`models.py`), comprising SimpleCNN, ConvNet4 and its functional forward pass; the meta-learning core (`maml.py`), comprising the inner adaptation loop, the first-order meta-update and the episodic evaluation routine; the three training entry points (`train_maml.py`, `train_baseline.py`, `train_pretrained.py`); the evaluation and figure-generation code (`evaluate.py`); and the supporting scripts.
- **Experiments.** All six training runs: SimpleCNN; ConvNet4 at 30 epochs; ConvNet4 tuned at 150 epochs; MAML 5-way 5-shot; and the two ResNet-18 runs under loss- and accuracy-based checkpointing. This includes the checkpoint-selection ablation of Section 3.2 and the per-class analysis of Section 3.
- **Writing.** The Method section (Section 2), including the formal treatment of MAML and the description of the architectures and training configuration, and the Results section (Section 3), including the interpretation of the baseline comparison and the few-shot protocols.

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- **Literature review.** The survey underpinning Section 1: the FER2013 benchmark and its published results, from the original competition through to current state of the art; the few-shot learning literature, covering both optimisation-based and metric-based approaches; and the augmentation literature. This review established the gap the report addresses, namely that existing few-shot evaluations of FER sample episodes from the training emotion set and therefore measure adaptation to unseen images rather than to unseen categories.
- **Figures.** Preparation of the figures and tables presenting the experimental results, including the confusion matrices, the training curves and the system overview diagram (Figure 1).
- **Manuscript review.** Review of the full manuscript for technical accuracy and consistency of the reported figures across sections.

The framing of the problem, the design of the held-out-emotion protocol and the choice of *disgust* and *surprise* as the withheld classes emerged from joint discussion between both authors, as did the decision to adopt first-order MAML. Both authors reviewed and approved the final manuscript.

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